

Body-fitted external flow validation

SIMPLE/Rhie-Chow; first order upwind; strict independent physical error <3%.

NACA reference: vector-digitized Di Ilio 2020, uncertainty +/-0.0001.

Historic NACA 0.205/0.120 pass claim is withdrawn.

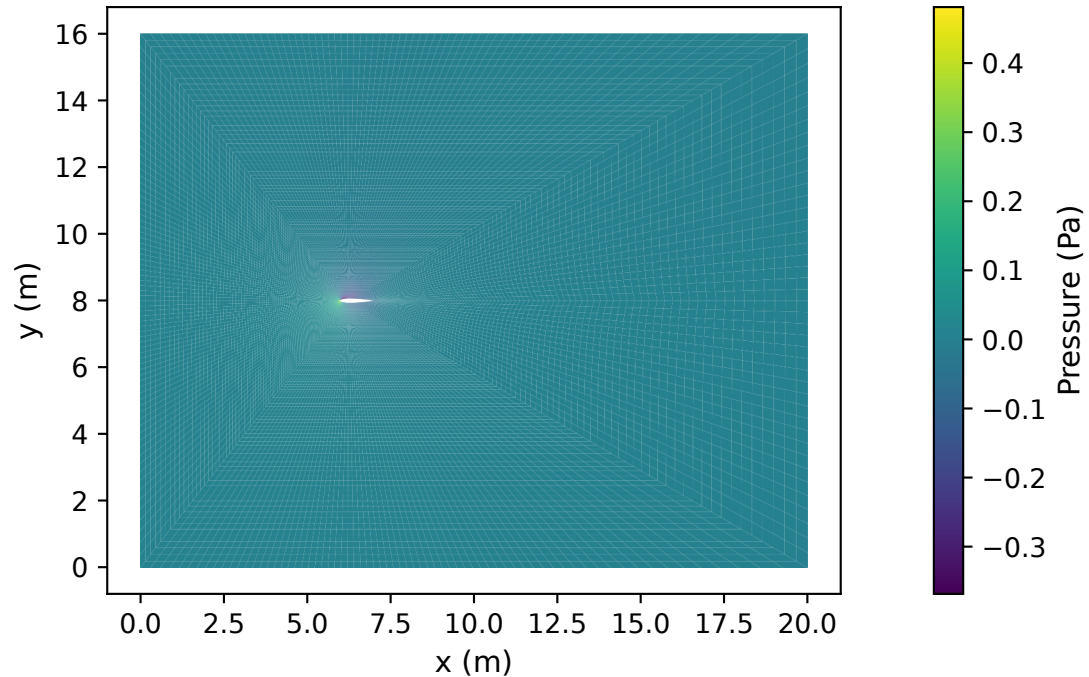
Single grid and residual convergence do not establish physical accuracy.

naca 256x104 cpu: converged=True, passed=False

{'drag': 0.14480159856236716, 'lift': 0.23290889750064211}

{'drag_relative_error': 0.1592795009448909, 'lift_relative_error': 0.15953758659414258, 'solver_residual': 9.998018005720932e-07, 'convergence_failure': 0}

naca 256x104



naca 256x104

